# ■ Astronomy

# Stars, Galaxies & Cosmic Scale

Grade 6 STEM Lesson

## ■ Learning Objectives

✔ Describe what stars are and how they produce light and heat.
✔ Identify and distinguish the main types of stars and galaxies.
✔ Explain the life cycle of a star from nebula to its final stage.
✔ Understand the concept of a light year and cosmic scale.
✔ Describe the methods humans use to explore space.

## ■ Section 1: What Are Stars?

Stars are giant balls of hot gas — mostly hydrogen and helium — that shine brightly in the sky. They produce light and heat through nuclear reactions deep inside their cores. Our own Sun is a star, just 150 million km away!

■ *Fun Fact: Every twinkle in the night sky is a giant ball of fire, millions of kilometres away!*

### Types of Stars

| Type | Description | Example |
|---|---|---|
| Yellow Dwarf | Medium-sized, steady light | Our Sun |
| Red Giant | Older, expanded star, cooler | Aldebaran |
| Blue Supergiant | Very large, extremely hot & bright | Rigel |
| White Dwarf | Remnant of a dead star, very dense | Sirius B |

## ■ Section 2: Life Cycle of a Star

| Stage | What Happens |
|---|---|

| | |
|---|---|
| **1. Nebula** | A cloud of gas and dust that begins to pull together under gravity. |
| **2. Protostar** | The collapsing cloud heats up to form a baby star. |
| **3. Main Sequence** | The stable, adult stage — our Sun is here now! Lasts billions of years. |
| **4. Red Giant** | The star expands as it runs low on hydrogen fuel. |
| **5. End Stage** | Small stars → White Dwarf. Massive stars → Supernova → Neutron Star or Black Hole. |

## ■ Section 3: Galaxies

A galaxy is a massive collection of stars, gas, and dust all bound together by gravity. Our Solar System lives inside the Milky Way — a spiral galaxy about 100,000 light years across, containing hundreds of billions of stars.

| Galaxy Type | Shape | Key Feature |
|---|---|---|
| Spiral | Swirling arms around a bright centre | Like the Milky Way — our home! |
| Elliptical | Round or oval, like a glowing egg | Contains mostly older, redder stars |
| Irregular | No defined shape — messy & unique | Often formed by galaxy collisions |

## ■ Section 4: Cosmic Scale & Light Years

Space is almost impossibly large. Astronomers use a special unit called a **light year** — the distance light travels in one year — to measure cosmic distances. Light travels at 300,000 km per second!

| Object | Size / Distance |
|---|---|
| Earth | 12,742 km diameter |
| Sun | 1.4 million km diameter (~109 Earths wide!) |
| Earth → Sun | 150 million km (8 light minutes) |
| Solar System | About 287 billion km across |
| Milky Way Galaxy | ~100,000 light years wide |
| Nearest galaxy (Andromeda) | ~2.5 million light years away |
| Observable Universe | ~93 billion light years across |

■ *Fun Analogy: If the Sun were a basketball, Earth would be a tiny pea placed 26 metres away!*

## ■ Section 5: How Do We Explore Space?

| Tool | What It Does |
| --- | --- |
| **Telescopes** | Collect light from far-away stars and galaxies; the James Webb Space Telescope can see billi |
| **Satellites** | Orbit Earth to collect data, take images, and measure radiation. |
| **Space Probes** | Robotic spacecraft sent to distant planets and beyond (e.g. Voyager 1, now outside our Solar |
| **Astronauts** | Humans who travel to space to conduct experiments — currently aboard the International Spa |

## ✏■ Review Questions

1. What is a star made of, and how does it produce energy?
2. Name three types of stars and one feature of each.
3. Describe the life cycle of a star like our Sun from birth to death.
4. What is a galaxy? How is the Milky Way different from an elliptical galaxy?
5. What is a light year? Why do scientists need this unit?
6. If the Sun were a basketball, what would Earth be, and how far away?
7. Name two tools astronomers use to study space and explain what each does.

> ■ *Did You Know? There are more stars in the observable universe than grains of sand on all of Earth's beaches combined. The universe is about 13.8 billion years old — and it's still expanding!*

## ■ Keep Exploring!

★ Go stargazing on a clear night — see how many stars you can count!
★ Visit a nearby planetarium or science museum.
★ Use a free app like Stellarium or Sky Map to identify stars.
★ Research India's ISRO space missions — Chandrayaan and Mangalyaan!
★ Read books about famous astronomers like Aryabhata, Galileo, and Carl Sagan.